Recap of last lecture

$$P^{VAP} \approx P^{SAT} \exp\left[\frac{\left(P - P^{SAT}\right)\underline{V}^{l}}{\mathrm{R}T}\right]$$

The Poynting correction

Fugacity

For a pure substance: $\underline{G}_i(T,P) = \underline{G}_i^0(T,P^0) + RT\ln\frac{f_i}{P^0}$

For a mixture: $\mu_i = \mu_i^0 + RT\ln\frac{\overline{f}_i}{P^0}$

$$\underline{G} = \sum \mu_i x_i = \sum \mu_i^0 x_i + RT\sum x_i \ln\frac{\overline{f}_i}{P^0}$$

Example of calculating fugacity

Using steam tables, calculate the fugacity of water at:

(a) 400°C and 20 bar

(b) 400°C and 500 bar

Critical temperature of water = 374ºC

Compare the results with the predictions of the Peng-Robinson EOS.

N.B. this is not an example of the Poynting correction - we are considering pure water vapour here.

Pure substance: $\underline{G}_i = \underline{G}_i^0 + RT \ln \frac{f_i}{P^0}$

Pure IG at P^0, T

$d\underline{G} = -\underline{S}dT + \underline{V}dP \Rightarrow RTd \ln f_i = \underline{V}_i dP$ (at constant T)

Start at $P = 0$ $\Rightarrow$ all gases ideal $\Rightarrow$ $f_i^{P=0} = P = 0$

$$RT \ln \frac{f_i}{f_i(P=0)} = \int_0^P \underline{V}_i dP$$

We just need $\underline{V}_i$ as a fn(P) to calculate f_i (or $\underline{G}_i$ or μ_i)

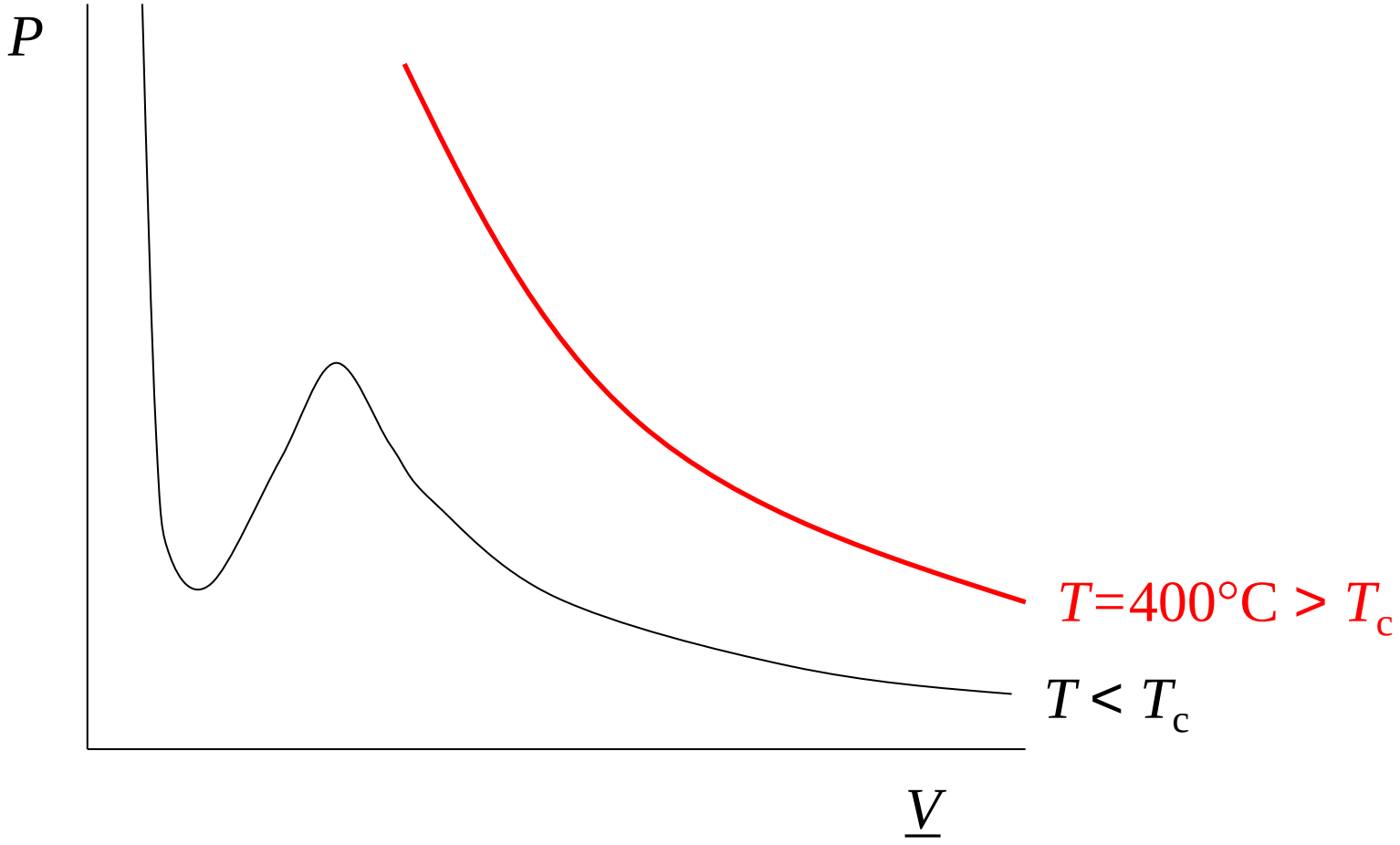
Use EOS or steam tables

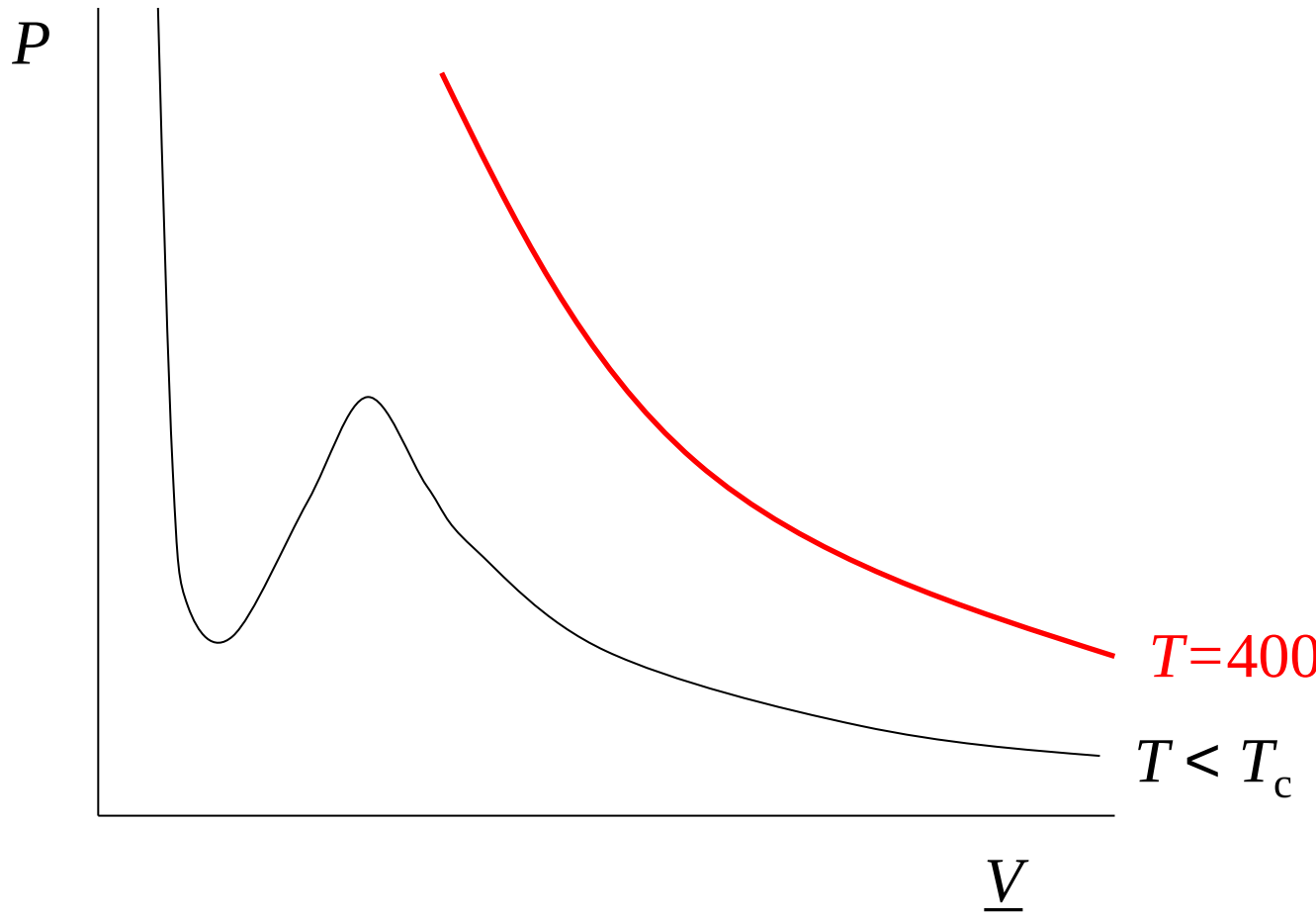
Data from steam tables in Sandler

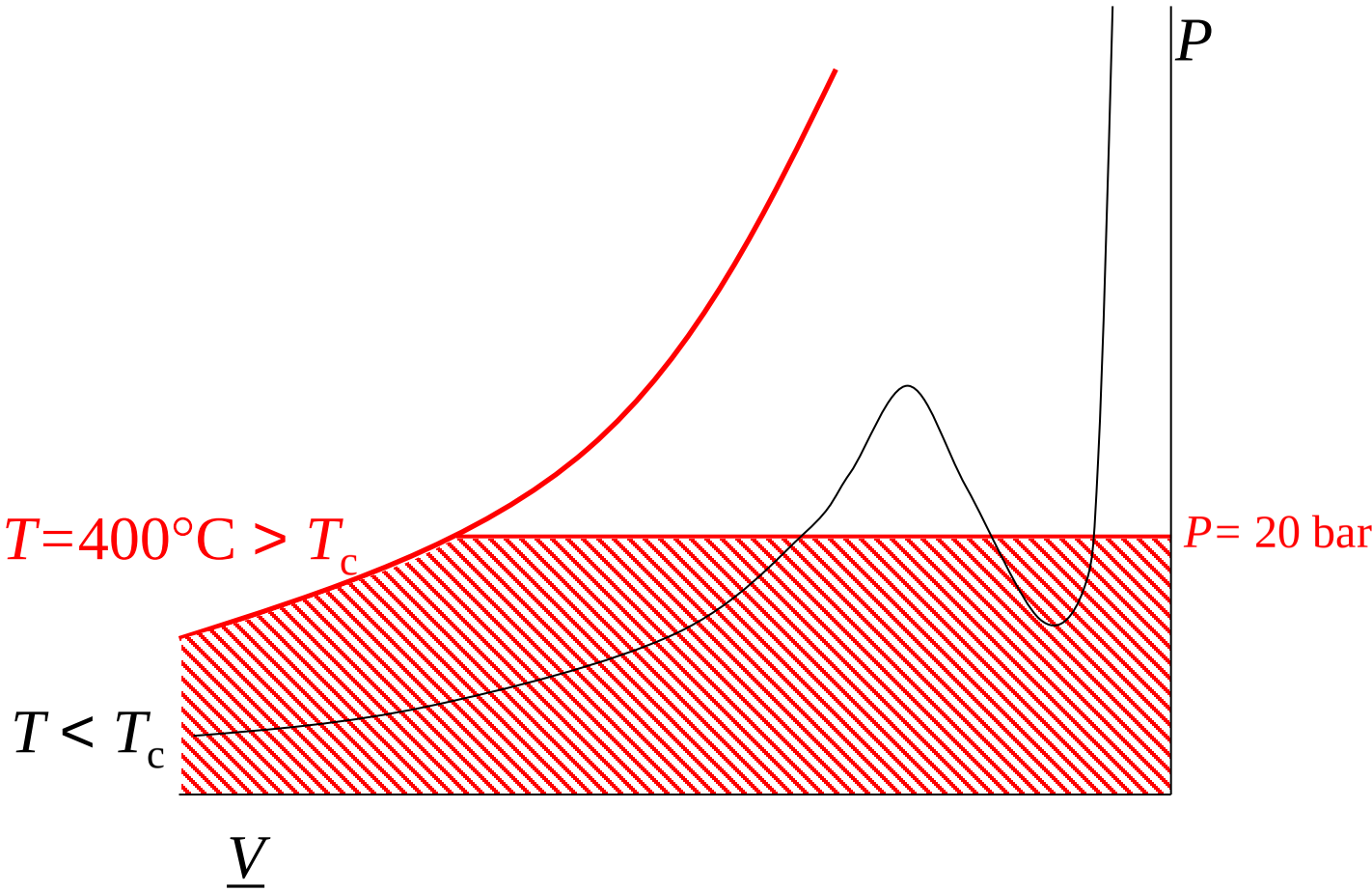
P (bar)	$\underline{V}$(m^3/mol) $\times 10^3$
0	∞
2	27.89
4	13.91
6	9.247
8	6.917
10	5.519
12	4.586
14	3.920
16	3.421
18	3.032
20	2.722

Small problem – we are trying to integrate from $\underline{V}_i = \infty$!

Plot water isotherm at 400°C:

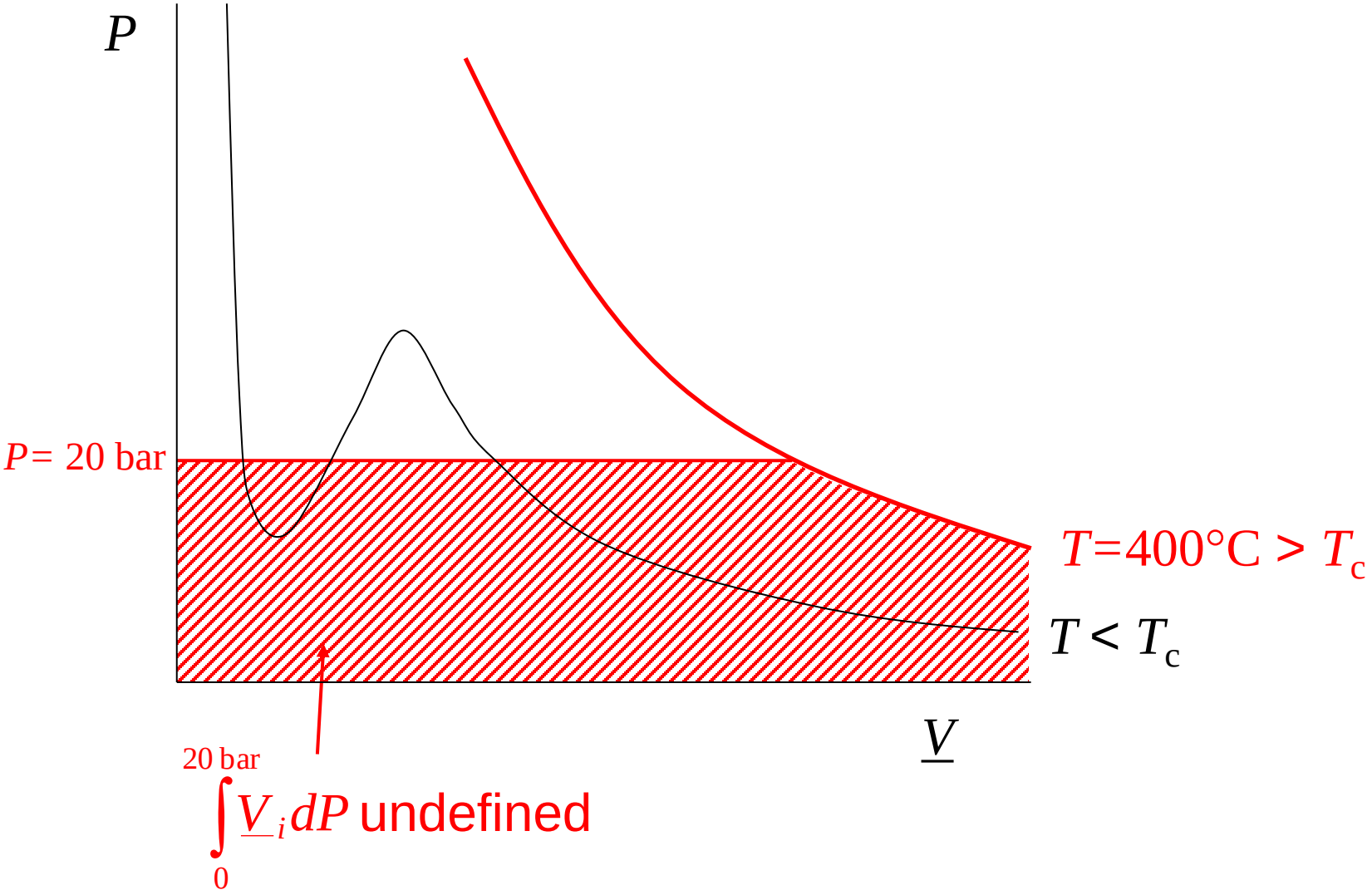



P
T=400
T < T_c
V


P
T=400°C > Tc
P= 20 bar
T < Tc
V

Small problem – we are trying to integrate from $\underline{V}_i = \infty$!

Plot water isotherm at 400°C:



A trick – calculate **deviation** from ideality along isotherm

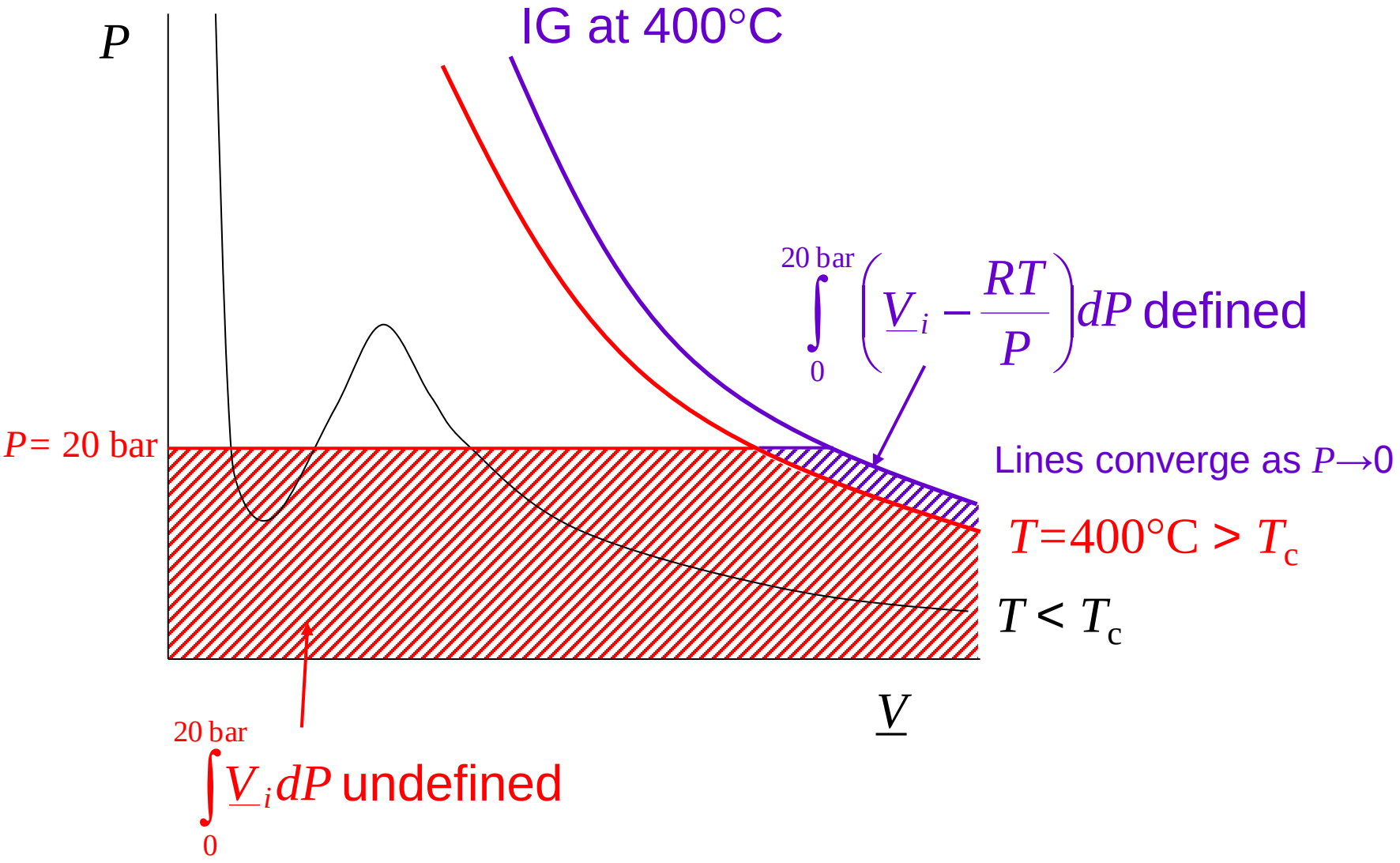
For IG $\underline{V}_i = \dfrac{RT}{P}$ and $f_i^{IG} = P$ (pure IG)

$$RT\left(\ln f_i - \ln f_i^{IG}\right) = RT \ln \frac{f_i}{P} = \int_0^P \left(\underline{V}_i - \frac{RT}{P}\right) dP$$

Define a new “fudge factor” (sorry),

$$\phi_i = \frac{f_i}{P}$$ fugacity coefficient

And, for mixtures $$\bar{\phi}_i = \frac{\bar{f}_i}{P_i}$$ fugacity coefficient


P
IG at 400°C
$\int_0^{20\ \text{bar}} \left(\underline{V}_i - \frac{RT}{P}\right) dP$ defined
P= 20 bar
Lines converge as $P \to 0$
$T = 400°\text{C} > T_c$
$T < T_c$
$\underline{V}$
20 bar
$\int_0^{20\ \text{bar}} \underline{V}_i\, dP$ undefined

P (bar)	$\underline{V}$ (m^3/mol) $\times 10^3$	RT/P (m^3/mol) $\times 10^3$	$\underline{V} - RT/P$ (m^3/mol) $\times 10^3$
0	∞	∞	0
2	27.89	27.98	-0.09
4	13.91	13.99	-0.08
6	9.247	9.328	-0.081
8	6.917	6.996	-0.079
10	5.519	5.597	-0.078
12	4.586	4.664	-0.078
14	3.920	3.998	-0.078
16	3.421	3.498	-0.077
18	3.032	3.109	-0.077
20	2.722	2.798	-0.076

(a) Trapezoidal rule:

$$\int_0^P \left(\underline{V}_{H_2O} - \frac{RT}{P} \right) dP = 2\times10^2 \left\{ \frac{0}{2} - 0.09 - .... - \frac{0.076}{2} \right\}$$

$$= -1.5\times10^2 \text{ J mol}^{-1}$$

$f_{H_2O}\left(400^\circ\text{C}, 20 \text{ bar}\right) = 19.4 \text{ bar}$ almost IG

$\varphi_{H_2O}\left(400^\circ\text{C}, 20 \text{ bar}\right) = 0.97$ P-R EOS $\Rightarrow \varphi_{H_2O} = 0.97$

(b) Calculate as above:

$f_{H_2O}\left(400^\circ\text{C}, 500 \text{ bar}\right) = 215 \text{ bar}$

$\varphi_{H_2O}\left(400^\circ\text{C}, 500 \text{ bar}\right) = 0.43$ P-R EOS $\Rightarrow \varphi_{H_2O} = 0.42$

High pressure $\Rightarrow$ high deviation from IG, but P-R is still good